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Original Research

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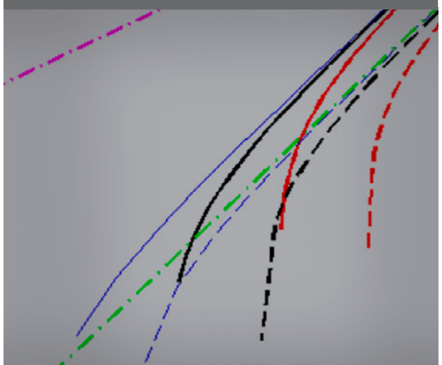

How does water flow from soil to roots? Theory suggests that root uptake from drying soils is controlled by the hydraulic properties of the soil near the roots, the rhizosphere. Recently, unexpected properties of the rhizosphere have been reported. We used a one-root model to investigate the effects of the rhizosphere on water availability to plants.

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How the Rhizosphere May Favor Water Availability to Roots

Recent studies have shown that rhizosphere hydraulic properties may differ from those of the bulk soil. Specifically, mucilage at the root–soil interface may increase the rhizosphere water holding capacity and hydraulic conductivity during drying. The goal of this study was to point out the implications of such altered rhizosphere hydraulic properties for soil–plant water relations. We addressed this problem through modeling based on a steady-rate approach. We calculated the water flow toward a single root assuming that the rhizosphere and bulk soil were two concentric cylinders having different hydraulic properties. Based on our previous experimental results, we assumed that the rhizosphere had higher water holding capacity and unsaturated conductivity than the bulk soil. The results showed that the water potential gradients in the rhizosphere were much smaller than in the bulk soil. The consequence is that the rhizosphere attenuated and delayed the drop in water potential in the vicinity of the root surface when the soil dried. This led to increased water availability to plants, as well as to higher effective conductivity under unsaturated conditions. The reasons were two: (i) thanks to the high unsaturated conductivity of the rhizosphere, the radius of water uptake was extended from the root to the rhizosphere surface; and (ii) thanks to the high soil water capacity of the rhizosphere, the water depletion in the bulk soil was compensated by water depletion in the rhizosphere. We conclude that under the assumed conditions, the rhizosphere works as an optimal hydraulic conductor and as a reservoir of water that can be taken up when water in the bulk soil becomes limiting.

How does water flow from soils to roots? Does the soil, in particular the soil in the immediate vicinity of roots, play a significant role in root water uptake? Due to the high root length density and the relatively high hydraulic resistance of roots, soil properties are not limiting for root uptake for a broad range of soil water potentials (Gardner, 1960). Below ground, the largest gradient in water potential occurs in the root tissue and across the root–soil interface, while the profiles of water potential and water content toward the roots remain generally quite flat. As the soil dries and the soil hydraulic conductivity falls, however, a critical drop in water potential may occur in the soil because of the radial geometry of the flow to roots and the nonlinearity of the unsaturated soil conductivity (Gardner, 1960; Jakobsen, 1974; de Willigen and van Noordwijk, 1987; de Jong van Lier et al., 2006; Schröder et al., 2008). In this case, the soil next to the roots may become a limiting factor for water flow in the belowground part of the soil–plant–atmosphere continuum. If the drying period continues, roots shrink and lose contact with the soil (Carminati et al., 2009). The formation of air-filled gaps at the soil–root interface will induce an additional resistance for water flow to the roots.

What are the strategies of plants to overcome water stress as the soil dries out? An answer to this question might come from several recent studies directly or indirectly addressing the hydraulic properties of the rhizosphere. The rhizosphere is the portion of the soil that is actively modified by roots as well as by microorganisms living in symbiosis with the roots. Root growth and root water uptake induce mechanical and hydraulic stresses that are expected to alter the density and structure of the rhizosphere. Mucilage exuded by the roots, with its high water holding capacity (McCully and Boyer, 1997), may increase the water holding capacity of the rhizosphere. On the other hand, lipids present in the mucilage may decrease the surface tension of the soil solution, decreasing the rhizosphere water holding capacity (Read et al., 2003). How do these modifications influence the water flow from soil to roots? Despite the large amount of literature reporting on the specific and distinct properties of the rhizosphere, as documented in the reviews by Gregory (2006) and Hinsinger et al. (2009), there are only a few studies investigating the effects of the rhizosphere's properties on root water uptake.

The goal of this study was to investigate the implications of common approaches neglecting rhizosphere properties in the study of plant–soil water relations. The study did not aim to answer a biological question but rather to point out the effects of the rhizosphere on water flow from soil to roots. This study stemmed from our previous observations of unexpected water dynamics in the rhizosphere. Carminati et al. (2010) observed larger water content in the rhizosphere than in the bulk soil during drying. Immediately after rewetting, the picture reversed and the rhizosphere remained drier than the bulk soil. During the following days, the water content of the rhizosphere increased, exceeding that of the bulk soil. These observations were explained by mucilage exuded by the roots. Mucilage is composed of polymeric substances, mainly sugars, having a high water holding capacity (McCully and Boyer, 1997; Read and Gregory, 1997; Read et al., 1999). After drying, the mucilage might become partly hydrophobic and rehydrate at a slower rate than the bulk soil. This would temporarily decrease the wettability of the rhizosphere. Such a dynamic and hysteretic behavior of the rhizosphere could explain parts of the contradictions in the studies on rhizosphere properties.

Currently, there are two concepts in the literature on the differences in hydraulic properties between the rhizosphere and the bulk soil: the water holding capacity of the rhizosphere is either higher or lower than in the bulk soil. The first case was documented in experiments by Young (1995) and Nakanishi et al. (2005) and in recent observations with nuclear magnetic resonance (S. Haber-Pohlmeier and A. Pohlmeier, personal communication, 2009), although these last observations could be partly explained by root hairs. The second case, which assumes lower water holding capacity of the rhizosphere, was discussed by Dunbabin et al. (2006) in a modeling study of water and nutrient uptake by roots. They justified the assumed rhizosphere properties by the presence of surfactant lipids in the root-exuded mucilage (Read et al., 2003). This assumption was also corroborated by observations of water repellency in the rhizosphere (Hallett et al., 2003; Whalley et al., 2004).

We believe that the two ideas are not in contradiction, but they rather reflect the dynamic and hysteretic hydraulic behavior of the rhizosphere. Depending on the state of hydration and chemical composition of the mucilage exuded by roots and bacteria, the rhizosphere may be in a "wet" or "dry" state, may not be in equilibrium with the bulk soil, and may exhibit different rates of wetting and drying. Our understanding of these complex processes is still developing, and in this study we decided to focus on one specific situation, when the soil dries and the rhizosphere holds more water than the bulk soil. We do not claim that this is a unique situation occurring in nature for all plant species, soil types, and atmospheric conditions. This situation is consistent, however, with important literature on mucilage (Young, 1995; McCully and Boyer, 1997) and recent observations with imaging techniques (Nakanishi et al., 2005; Tumlinson et al., 2008; Carminati et al., 2010), and we think that it represents a common case for water uptake by roots in drying soils.

The specific goal of this study was to investigate how the supposed rhizosphere properties affect root water uptake. We hypothesized that the rhizosphere holds more water than the bulk soil at the same negative potential during the drying phase. We also assumed that the hydraulic conductivity in the rhizosphere decreases more slowly than in the bulk soil. These assumptions were corroborated by studies on soils amended with extracellular polymeric substances (EPS), which we expected to mimic the effects of mucilage on rhizosphere properties. Chenu and Roberson (1996) found that EPS increases the diffusion of glucose under unsaturated conditions. The higher diffusion rates of glucose can be explained by the capacity of the EPS network to remain saturated at very negative potentials (Or et al., 2007). We simplified the problem by assuming that a root is a cylinder of infinite length, and the rhizosphere and the soil domain are concentric cylinders with different hydraulic properties. For simplification, we neglected day–night flow rate variations and assumed a constant uptake rate. For such conditions, we calculated the water potentials toward a root by implementing the steady-rate analytical approach used in several studies (de Willigen and van Noordwijk, 1987; de Jong van Lier et al., 2006; Schröder et al., 2008; Schneider et al., 2010). The analytical approach was validated by comparison with numerical simulations.

The present study goes beyond the results of Carminati et al. (2010), in which the modeling approach was just shortly illustrated. In particular, in Carminati et al. (2010) the capacity of the rhizosphere to release more water than the bulk soil and thus attenuate the water potential gradients in the bulk soil was not properly described. In the present study, we also evaluated the effects of the rhizosphere on the effective hydraulic conductivity of the soil–rhizosphere–plant continuum.

Materials and Methods

Root water uptake was modeled assuming that the root, rhizosphere, and bulk soil are concentric cylinders of radii r_0, r_1, and r_2, respectively (Fig. 1). The hydraulic properties of the rhizosphere and the bulk soil were taken from previous experiments with neutron radiography (Carminati et al., 2010), where we derived the water retention curve of the rhizosphere of a plant growing in a sandy soil. For the unsaturated conductivity of the rhizosphere, we referred to studies on porous media containing EPS (Chenu and Roberson, 1996). Because our hypothesis was that the high water holding capacity of the rhizosphere is primarily caused by root-exuded mucilage and because mucilage is mostly composed of polymeric substances, we expected that the hydraulic properties of soil with EPS would mimic those of the rhizosphere. According to Or et al. (2007), EPS reduce the saturated soil conductivity by a few orders of magnitude but increase the unsaturated conductivity. The hydraulic properties of the rhizosphere and bulk soil were parameterized according to Brooks and Corey (1964) with the following notation:

$$\Theta = \left(h/h_0\right)^{-\lambda} \quad [1]$$

and

$$k = k_{\mathrm{sat}}\left(h/h_0\right)^{-\tau} \quad [2]$$

where Θ is the water saturation, defined as $\Theta = (\theta - \theta_r)/(\theta_s - \theta_r)$, θ is the volumetric water content (cm^3 cm^{-3}), θ_r is the residual water content, θ_s is the water content at saturation, h is the matric head, which is the matric potential expressed as a hydraulic head (cm), h_0 is the air-entry value (cm), k is the hydraulic conductivity (cm s^{-1}), k_{sat} is the conductivity at saturation, and λ and τ are fitting parameters. The functions $h(\theta)$ and $k(h)$ of the bulk soil and rhizosphere are plotted in Fig. 1. The parameters of the hydraulic functions are given in Table 1. In this study, we limited our analysis to these soil properties.

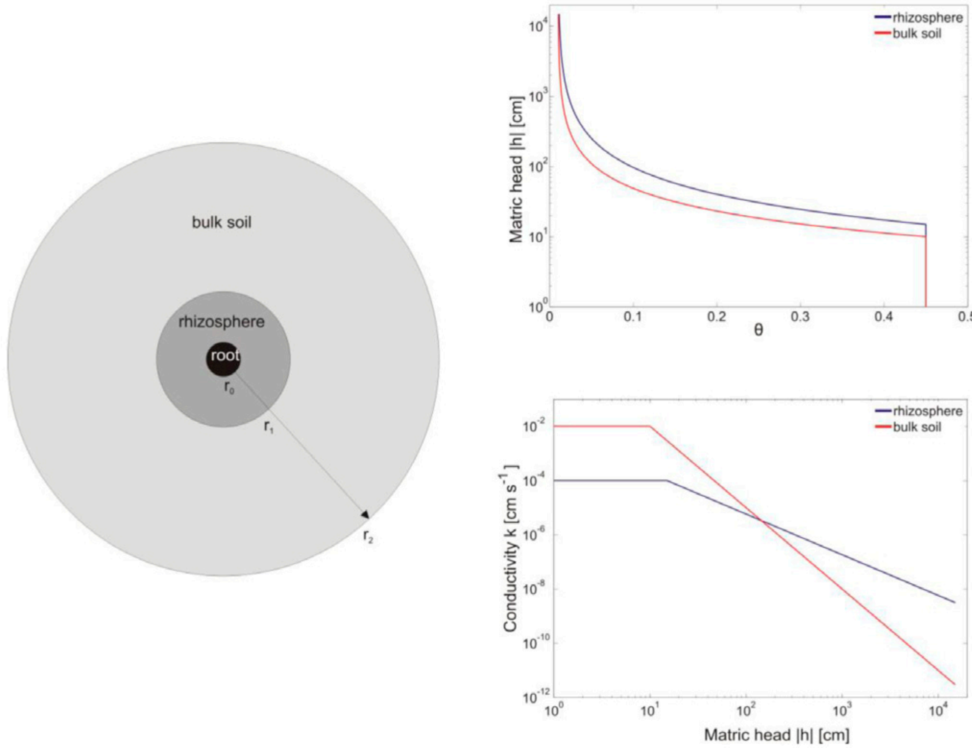

Fig. 1. Scheme of the model domain, showing the bulk soil, rhizosphere, and root as concentric circles (left), and water retention curves (top right) and unsaturated conductivities (bottom right) of the rhizosphere and bulk soil, as estimated by Carminati et al. (2010).

The water flow to the root was calculated solving the Richards equation in radial coordinates:

$$\frac{\partial \theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[r k(h)\frac{\partial h}{\partial r}\right] \quad [3]$$

where t is time and r is the radial coordinate.

Equation [3] was linearized using the matric flux potential ϕ (m^2 s^{-1}):

$$\phi(h) = \int_{h_{\mathrm{lim}}}^{h} k(h')\,\mathrm{d}h' \quad [4]$$

where h_{lim} is a reference matric potential that we set equal to the permanent wilting point, $h_{\mathrm{lim}} = -15{,}000$ cm. Then Eq. [3] transforms to

$$\frac{\partial \theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial \phi}{\partial r}\right] \quad [5]$$

Equation [5] was solved analytically under the assumption that $\partial\theta/\partial t$ is constant, i.e., it is not a function of r. This assumption is called a *steady-rate assumption* and it has been tested and discussed in several studies (Cowan, 1965; Jakobsen, 1974; de Willigen and van Noordwijk, 1987; de Jong van Lier et al., 2006; Raats, 2007; Schröder et al., 2008; Schneider et al., 2010). Under this assumption, the radial profile of the matric flux potential at each time step can be expressed as

$$\phi(r) = ar^2 + b\ln(r) + c \quad [6]$$

where a, b, and c are parameters that may vary with time (de Willigen and van Noordwijk, 1987). Substitution of Eq. [6] in Eq. [5] shows that $a = (1/4)(\partial\theta/\partial t)$. The determination of b and c is given below. Note that a, b, and c change with time, so that $\phi = \phi(r,t)$.

We started from the steady-rate assumption and we implemented the solution for the case of the two domains illustrated in Fig. 1. We assumed that the steady-rate assumption was satisfied in both domains, the bulk soil and the rhizosphere. As a convention, we use the subscripts 1 and 2 to refer to the rhizosphere ($i = 1$) and the bulk soil ($i = 2$). Then, $\phi_1(r,t)$ and $\phi_2(r,t)$ are the matric flux potentials in the two domains according to Eq. [6], and a_i, b_i, and c_i, with $i = 1$ and 2, are the respective parameters to be calculated for each time step. Defining $q(r,t)$ as the Darcy water flux den-

Table 1. The Brooks-Corey parameters of residual and saturated volumetric water content (θ_r and θ_s, respectively), air-entry value (h_0), hydraulic conductivity at saturation (k_{sat}), and fitting parameters λ and τ of the bulk soil and the rhizosphere.

Soil	θ_r	θ_s	h_0	λ	k_{sat}	τ
			cm		cm s^{-1}	
Bulk soil	0.01	0.45	−10	1	10^{-2}	3
Rhizosphere	0.01	0.45	−15	0.85	10^{-4}	1.5

sity (cm s^{-1}), then the two solutions have to satisfy the following boundary conditions:

$$q(r_0,t)=q_0 \quad [7a]$$

$$q(r_2,t)=0 \quad [7b]$$

$$h_1(r_1,t)=h_2(r_1,t) \quad [7c]$$

$$q(r_1,t)=-k_1(h_1)\left.\frac{\partial h_1}{\partial r}\right|_{r_1}=-k_2(h_2)\left.\frac{\partial h_2}{\partial r}\right|_{r_1} \quad [7d]$$

$$h(r_1,t=0)=h_{\text{init}} \quad [7e]$$

where r_0 is the root radius, q_0 is the flow at the root surface, and h_{init} is the initial water potential. Equation [7a] imposes a fixed flow equal to q_0 at the root surface, until the potential at the root surface, $h(r_0,t)$, reaches a limiting point. Here, the limiting point was set equal to h_{lim} = −15,000 cm, as in de Jong van Lier et al. (2006). Equation [7b] means no flow at the outer soil-cylinder radius. Equations [7c] and [7d] impose continuity of the matric potential and water flux at the boundary between the bulk soil and the rhizosphere. Equation [7e] gives the initial condition. It was sufficient to set the initial condition at a single point to calculate $h(r,t = 0)$ according to Eq. [6] and [7a–7e].

A note for solving Eq. [7a–0e]: by setting the initial potential at r_1, Eq. [7a–7e] become a system of linear equations for the parameters a_i, b_i, and c_i to be solved for each time t. The mathematical problem consists in determining these six parameters. Explicit derivation of the parameters is given in Appendix A. Determination of the six parameters requires a sixth boundary condition that accounts for the change in h with time. To account for continuity of the matric potential, the temporal change of h at r_1 has to be continuous across the interface between the bulk soil and the rhizosphere. This condition should be an automatic consequence of Eq. [7c]. Because the matric flux potential ϕ expressed as in Eq. [6] is not explicitly function of time, however, the continuity of the derivative of h with time has to be imposed. This leads to

$$\left.\frac{\partial h_1}{\partial t}\right|_{r_1}=\left.\frac{\partial h_2}{\partial t}\right|_{r_1} \quad [7f]$$

Because $\partial\theta/\partial t = 4a$, Eq. [7f] turns into

$$\frac{a_1}{C_1[h_1(r_1,t)]}=\frac{a_2}{C_2[h_2(r_1,t)]} \quad [8]$$

where $C(h) = \partial\theta/\partial h$ is the specific soil water capacity (cm^{-1}). Equation [8] determines the ratio of water that is taken up from the soil and rhizosphere. The specific soil water capacities $C(h)$ of the two domains at r_1 determine the ratio of the two parameters a_1 and a_2. This provides the sixth boundary condition needed to solve Eq. [6] for both domains i = 1 and 2.

We calculated the matric potential and water content profiles for various uptake rates, q_0 = 0.1, 0.5, 1, 1.5, and 2 cm d^{-1}, and external radii r_2 = 0.5, 1, 2, 3, and 5 cm. The root radius was set to r_0 = 0.05 cm and the rhizosphere radius was set to r_1 = 0.25 cm, as observed by neutron radiography (Carminati et al., 2010). As an initial condition, we set h_{init} = −20 cm. The water content $\theta(r,t)$ and the matric potential $h(r,t)$ were then computed for $t \leq t_{\text{lim}}$, where t_{lim} is the time when the matric potential at the root surface reaches the limiting matric potential, i.e., $h(r_0,t_{\text{lim}}) = h_{\text{lim}}$ = −15,000 cm.

The solutions were calculated for two cases: (i) the soil domain is composed of rhizosphere and bulk soil with distinct hydraulic properties; and (ii) the hydraulic properties of the rhizosphere and those of the bulk soil are identical. As an abbreviation, we refer to the first case as the case with a rhizosphere and to the second as the case with no rhizosphere.

Additionally, we addressed the computational question of whether the role of the rhizosphere can be simulated by extending the effective radius of the roots. To this end, we simulated the case of a root having a radius equal to r_1 and no rhizosphere.

Finally, we validated the proposed steady-rate analytical solution by comparison with a numerical simulation. The numerical problem was solved based on the scheme described in de Jong van Lier et al. (2006). Description of the numerical model and comparison with the analytical solution are provided in Appendix B.

Results and Discussion

In Fig. 2, we plotted the water content profiles calculated for the case with no rhizosphere (left) and the case with a rhizosphere (right), starting from h_{init} = −20 cm to the limiting point, i.e., $h(r_0)$ = −15,000 cm. The profiles are for q_0 = 0.5 cm d^{-1} and r_2 = 1 cm. With no rhizosphere, the water content was almost horizontal for all the time steps. A closer look would show that small gradients in water content profiles developed close to the roots as the soil dried. In the second case (Fig. 2, right), the rhizosphere was initially much wetter than the bulk soil. As the soil dried, the rhizosphere became almost as dry as the bulk soil. Small gradients in water content developed outside the rhizosphere as the soil became very dry (after 4 d and 13 h).

In Fig. 3, we plotted the matric potential profiles toward the root. With no rhizosphere (Fig. 3, left), the matric potential remained flat for the first 4 d. During the fifth day, matric potential gradients developed close to the root. After 4 d and 9 h, the matric potential dropped at the root surface and $h(r_0)$ reached the limiting point of −15,000 cm. At the same time, the matric potential at the outer radius was significantly higher, $h(r_2,t_{\text{lim}})$ = −2588 cm. With a rhizosphere (Fig. 3, right), the drop in matric potential was attenuated. After 4 d and 9 h, the matric potential was still relatively high, $h(r_2)$ = −550 cm, and almost horizontal. The limiting point at the root surface was reached at t_{lim} = 4 d and 13 h, 4 h later than in the case

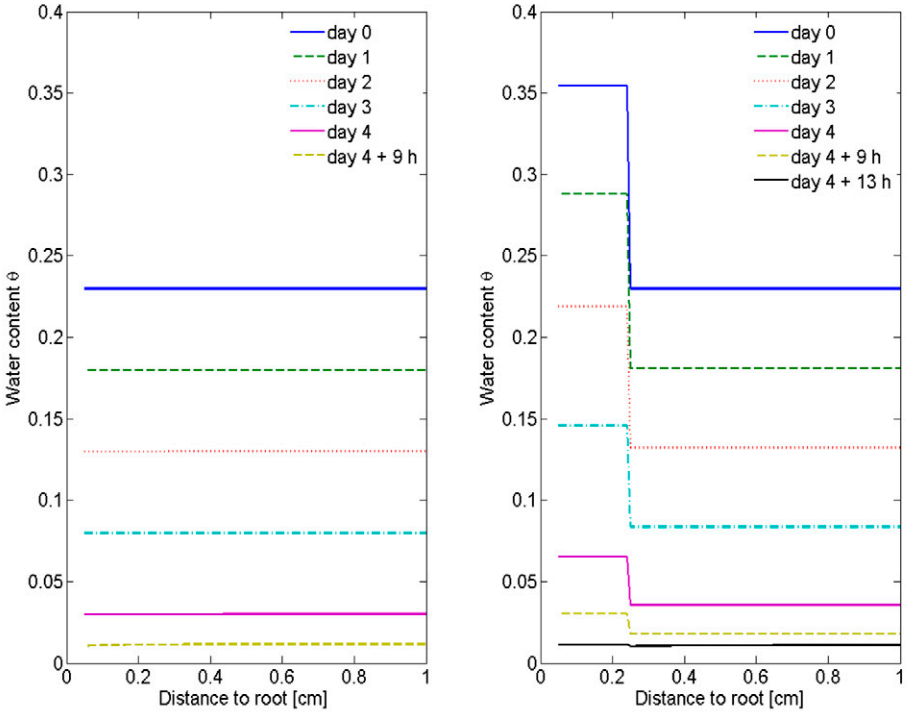

Fig. 2. Water content profiles toward a root in the case of homogeneous soil with no rhizosphere (left) and with a rhizosphere (right). The profiles were calculated for constant uptake q_0 = 0.5 cm d^{-1} and a bulk soil radius r_2 = 1 cm. The higher water content in the rhizosphere compared with the bulk soil was caused by the different water retention curves of the two domains. A close-up of the water content profile at the limiting point would show a slight decrease in water content toward the root (for the case with no rhizosphere) and just outside the rhizosphere (for the case with a rhizosphere).

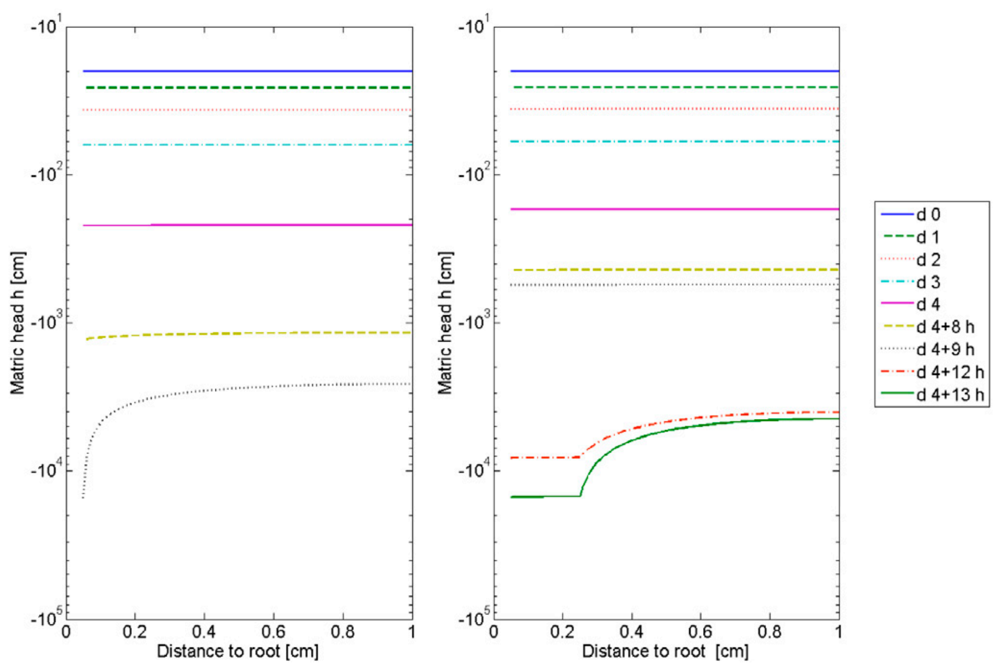

Fig. 3. Profiles of the matric potential $h(r,t)$, with h plotted vs. distance to a root r at different time steps t for the case with no rhizosphere (left) and with rhizosphere (right). The calculations correspond to the plots of water content shown in Fig. 2. The matric potential $h(r,t)$ was calculated to $h(r_0,t_{\text{lim}}) = -15{,}000$ cm.

with no rhizosphere. The matric potential at the outer radius at t_{lim} was approximately $h(r_2,t_{\text{lim}}) = -4447$ cm.

Figure 3 shows that the rhizosphere reduced the drop in matric potential toward the roots. This is because, at low matric potentials, the unsaturated conductivity of the rhizosphere is higher than that of the bulk soil and, hence, the gradients in matric potential are smaller within the rhizosphere. Specifically, the largest gradient in matric potentials occurred in the bulk soil at the border with the rhizosphere, i.e., $\max[\partial h(r,t_{\text{lim}})/\partial r] = [\partial h(r,t_{\text{lim}})/\partial r]|_{r1}$.

The fact that the largest matric potential gradients occurred at the rhizosphere radius and not at the smaller root radius implies a lower decrease in matric potential from the bulk soil to the root. The reason is the following: assuming a steady-state condition, the flow velocity $q(r)$ and the gradients $\partial h/\partial r$ are proportional to $1/r$. Therefore, the occurrence of the largest gradients at $r_1 > r_0$ means a smaller overall decrease in matric potential from r_2 to r_0. Thus, the effect of the rhizosphere would be similar to extending the root radius from r_0 to r_1, assuming that the matric potential gradients within the rhizosphere are negligible.

There is a second reason, however, why the drop in matric potential in the bulk soil was reduced due to the rhizosphere. The reason is illustrated in Fig. 4, where we plotted the changes in water content, $-\partial\theta/\partial t$, in the bulk soil and the rhizosphere. The results are compared with the changes in soil water content in the case with no rhizosphere. With no rhizosphere, $\partial\theta/\partial t$ was obviously constant because of the fixed flow boundary conditions. Including the rhizosphere, water was preferentially taken up from there. In this way, the hydraulic stress in the bulk soil was partly reduced and the drop in matric potential in the bulk soil was then attenuated. The behavior of $\partial\theta/\partial t$

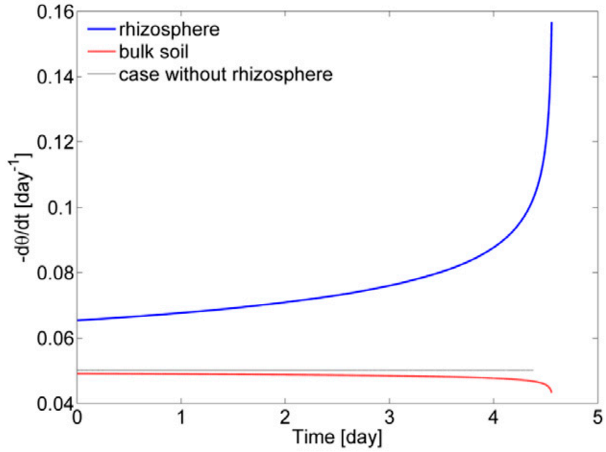

Fig. 4. Decrease in water content with time ($-d\theta/dt$) in the rhizosphere, in the bulk soil, and in the case with no rhizosphere.

in the rhizosphere and bulk soil can be explained by Eq. [8], which distributes the water uptake according to the specific soil water capacity of the two domains. In principle, if the bulk soil capacity was larger than that of the rhizosphere, more water would be depleted from the bulk soil than from the rhizosphere.

In Fig. 5, we show the profiles of matric potential h (left) and water content θ (right) at $t = t_{\text{lim}}$ for various uptake rates (q_0 = 0.1, 0.5, and 2 cm d^{-1}). The calculated t_{lim} for increasing q_0 were 22 d and 2 h, 4 d and 9 h, and 1 d and 2 h, respectively, for the case with no rhizosphere, and 22 d and 20 h, 4 d and 13 h, and 1 d and 3 h, respectively, for the case with a rhizosphere. The drop in h toward the root increased with increasing flow rates. Thus, the limiting point for water uptake was reached at a higher water content in the bulk soil, meaning that the efficiency of water use was considerably decreased. This effect was much less pronounced in the presence of a rhizosphere, where more water could be extracted from the bulk soil before the plant reached the limiting point.

To quantify the impact of the rhizosphere on water use efficiency, we calculated the volume of water extracted per unit of root length from $t = 0$ to $t = t_{\text{lim}}$ as $\Delta V_{\text{w}} = V_{\text{w}}(t_{\text{lim}}) - V_{\text{w}}(t = 0)$ (m^3 m^{-1}), where $V_{\text{w}}(t)$ (m^3 m^{-1}) is the volume of water in the sample per unit root length at time t. We were in the interested in the difference between ΔV_{w} calculated for the case with a rhizosphere, called $\Delta V_{\text{w}}^{\text{rhizo}}$, and ΔV_{w} calculated for the case with no rhizosphere, $\Delta V_{\text{w}}^{\text{no-rhizo}}$. Similarly, we define $t_{\text{lim}}^{\text{rhizo}}$ as t_{lim} in the case with a rhizosphere and $t_{\text{lim}}^{\text{no-rhizo}}$ for the case with no rhizosphere. The calculations were computed for varying the uptake rate q_0 for a fixed bulk soil radius r_2 = 1 cm (Fig. 6a, 6c, and 6e) and for varying r_2 for a fixed q_0 = 1 cm d^{-1} (Fig. 6b, 6d, and 6f). The root and rhizosphere radii were kept fixed for all scenarios, r_0 = 0.05 cm and r_1 = 0.25 cm. In all cases, we started from an initial matric potential of $h(r_1, t = 0) = -20$ cm.

In Fig. 6a and 6b, we plotted $\Delta V_{\text{w}}^{\text{rhizo}} - \Delta V_{\text{w}}^{\text{no-rhizo}}$ (m^3 m^{-1}). Figure 6a shows that $\Delta V_{\text{w}}^{\text{rhizo}} - \Delta V_{\text{w}}^{\text{no-rhizo}}$ increased with increasing flow rate, indicating that the effect of the rhizosphere was more significant when the flow rate was high. The increase in $\Delta V_{\text{w}}^{\text{rhizo}} - \Delta V_{\text{w}}^{\text{no-rhizo}}$ with increasing bulk soil radius r_2 (Fig. 6b) was caused by the increase in bulk soil volume per root length, $V_{\text{tot}} = \pi(r_2^2 - r_1^2)$ (m^3 m^{-1}).

In Fig. 6c and 6d, we plotted $\Delta V_{\text{w}}^{\text{rhizo}}/V_{\text{tot}} - \Delta V_{\text{w}}^{\text{no-rhizo}}/V_{\text{tot}}$ (blue circles). This is the difference in water volume extracted up to $t = t_{\text{lim}}$ per total volume between the cases with and without a rhizosphere. The increase in the available water content due to the rhizosphere was caused by two reasons: (i) thanks to the rhizosphere, the final water content in the bulk soil domain was slightly lower than in the case with no rhizosphere; and (ii) the initial total water

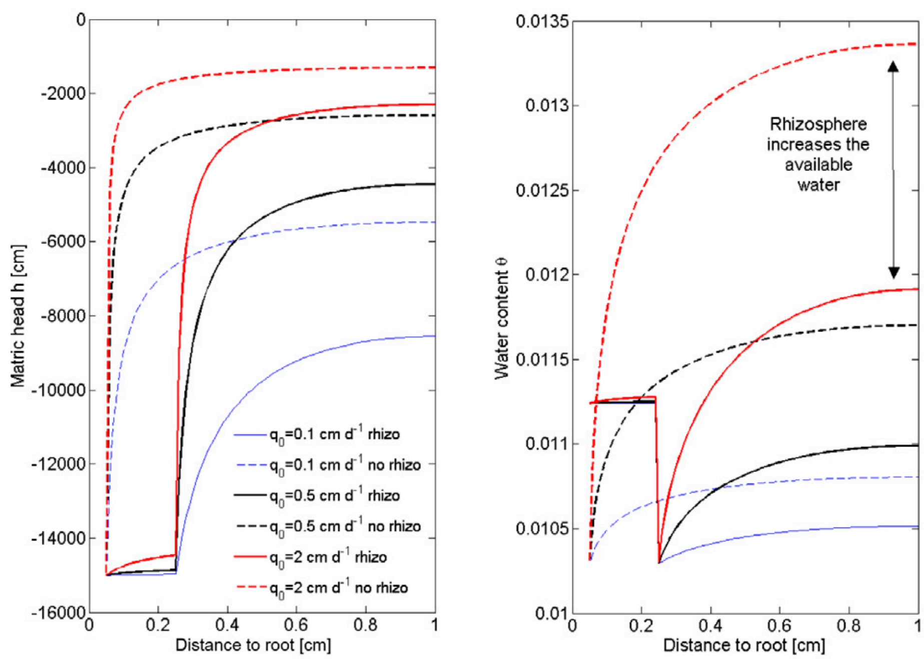

Fig. 5. Profiles of matric potential h (left) and water content θ (right) at time $t = t_{\text{lim}}$, where t_{lim} is the time at which the matric potential at the root surface reaches the limiting matric potential, here set to −15,000 cm. The profiles were calculated for low, medium, and high flow rates (q_0 = 0.1, 0.5, and 2 cm d^{-1}) and bulk soil radius r_2 = 1 cm for the case with a rhizosphere (solid line) and the case with no rhizosphere (dashed line). The calculated t_{lim} for increasing q_0 were 22 d and 2 h, 4 d and 9 h, and 1 d and 2 h for the case with no rhizosphere; and 22 d and 20 h, 4 d and 13 h, and 1 d and 3 h for the case with a rhizosphere.

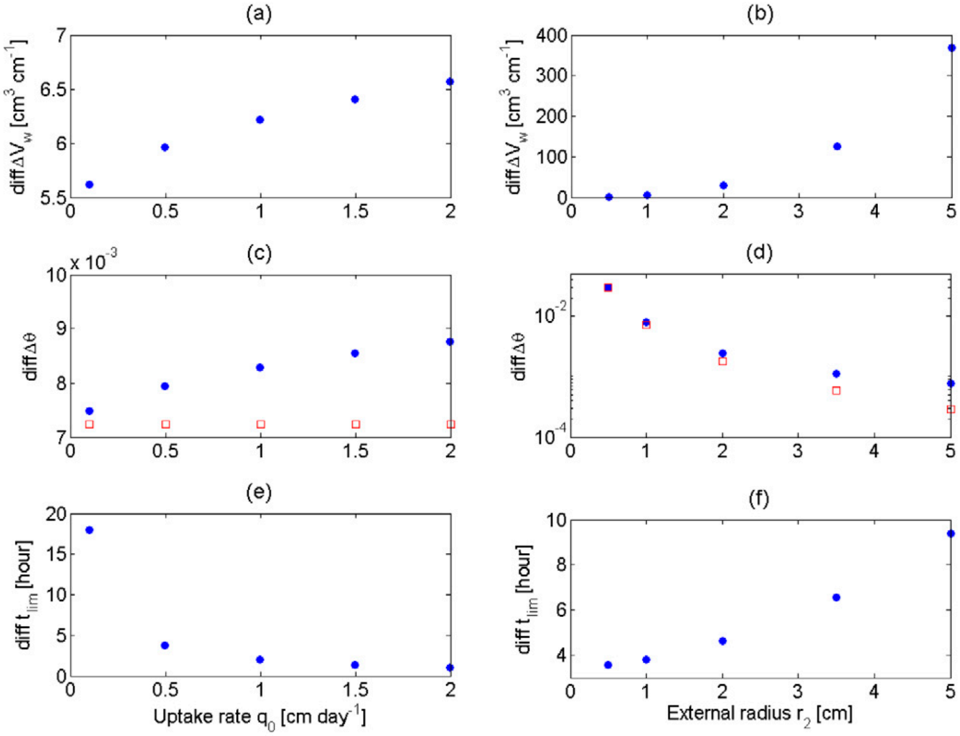

Fig. 6. Differences in the volume of water available to roots before the limiting point is reached, calculated for (a, c, and e) varying uptake rates q_0 with a fixed bulk soil radius r_2 = 1 cm and (b, d, and f) varying r_2 with a fixed q_0 = 1 cm d^{-1}: (a, b) volumes of water per root length taken up to the limiting point, ΔV_w (m^3 m^{-1}), showing the difference between the cases with a rhizosphere and with no rhizosphere, $\Delta V_w^{\text{rhizo}} - \Delta V_w^{\text{no-rhizo}}$; (c, d) difference in available water content with and without a rhizosphere, $\Delta V_w^{\text{rhizo}}/V_{\text{tot}} - \Delta V_w^{\text{no-rhizo}}/V_{\text{tot}}$, where V_{tot} is the volume of the domain per root length, with the red squares showing the differences in initial water content between the two cases; and (e, f) the difference between the time at which the matric potential at the root surface reaches the limiting matric potential, here set to −15,000 cm, with a rhizosphere and with no rhizosphere, $t_{\text{lim}}^{\text{rhizo}} - t_{\text{lim}}^{\text{no-rhizo}}$.

content in the case with a rhizosphere was higher than in the latter case (Fig. 2). To distinguish these two aspects, we plotted $V_w^{\text{rhizo}}(t = 0)/V_{\text{tot}} - V_w^{\text{no-rhizo}}(t = 0)/V_{\text{tot}}$ (Fig. 6c, red squares). This is the difference between the initial average water content in the cases with and without a rhizosphere. The calculated difference in the initial water content was obviously constant for different uptake rates q_0 (Fig. 6c). It is also obvious that the difference decreased with increasing r_2, because the rhizosphere volume relative to the total volume decreased with increasing r_2 (Fig. 6d). The plotted data show that the difference in the initial water content made the main contribution to the total $\Delta V_w^{\text{rhizo}}/V_{\text{tot}} - \Delta V_w^{\text{no-rhizo}}/V_{\text{tot}}$. For increasing q_0 and r_2, however, the importance of the average water content at the limiting point became more and more significant.

In Fig. 6e and 6f, the effect of the rhizosphere on the time to reach the limiting point is shown in terms of $t_{\text{lim}}^{\text{rhizo}} - t_{\text{lim}}^{\text{no-rhizo}}$. The nonlinear, convex decrease of $t_{\text{lim}}^{\text{rhizo}} - t_{\text{lim}}^{\text{no-rhizo}}$ with flow rate indicates that the effect of the rhizosphere on delaying t_{lim} was more significant for high transpiration rates. If the effect of the rhizosphere had not been dependent on q_0, $t_{\text{lim}}^{\text{rhizo}} - t_{\text{lim}}^{\text{no-rhizo}}$ would have decreased linearly with increasing q_0. The time gained thanks to the rhizosphere was on the order of some hours. Such a delay in t_{lim} could be significant when considering that overnight the soil can be remoistened by water redistribution from deeper soil regions, hydraulic lift, and changing atmospheric conditions. The increase in $t_{\text{lim}}^{\text{rhizo}} - t_{\text{lim}}^{\text{no-rhizo}}$ with increasing r_2 (Fig. 6f) shows that the positive effect of the rhizosphere in increasing the available water became more significant for low root length density distributions.

Above we quantified how the rhizosphere increases the availability of water to roots. In this context, a relevant aspect is the matric potential gradient required to drive water from the bulk soil toward the root. This is the matric potential lost due to viscous dissipation in the soil. To illustrate this aspect, we plotted the matric potential at the root surface, $h(r_0,t)$, as a function of the matric potential at the outer radius of the domain, $h(r_2,t)$, for various uptake rates, q_0, with and without a rhizosphere (Fig. 7). For $h(r_2,t) > -500$ cm, $h(r_0,t)$ is approximately equal to $h(r_2,t)$. This means negligible dissipation due to viscosity and small matric potential gradients across the soil. For $h(r_2,t) < -500$ cm, $h(r_0,t)$ starts to deviate from $h(r_2,t)$ and becomes much more negative. At the limiting point, the slope of the curve becomes extremely steep. This is the point where the optimal transpiration demand cannot be sustained anymore. The decrease in $h(r_0,t)$ is more accentuated for high q_0. In all cases, the effect of the rhizosphere is to delay and attenuate the decrease in $h(r_0,t)$ and therefore reduce the energy dissipation across the soil.

The difference in matric potential across r_2 and r_0 is related to the effective conductivity of the flow domain, $k_{\text{eff}}(h)$. To calculate $k_{\text{eff}}(h)$, we assumed that the soil conductivity is not dependent on the radial coordinate r. This assumption is valid as long as the changes in the hydraulic conductivity toward the roots are small, i.e., $k[h(r_0,t)] \sim k[h(r_2,t)]$, or, in other words, when the matric potential gradients toward a root are small. In this case and for steady-rate conditions, $k_{\text{eff}}(h)$ is given by

$$k_{\text{eff}} = -\frac{r_0\, q_0 \left\{\ln\left(r_2/r_0\right)\left[r_2^2/r_2^2 - r_0^2\right] - \left(1/2\right)\right\}}{h\left(r_2\right) - h\left(r_0\right)} \qquad [9]$$

Note that Eq. [9] gives the actual soil conductivity only in the case of uniform conductivity. The derivation of Eq. [9] is given in

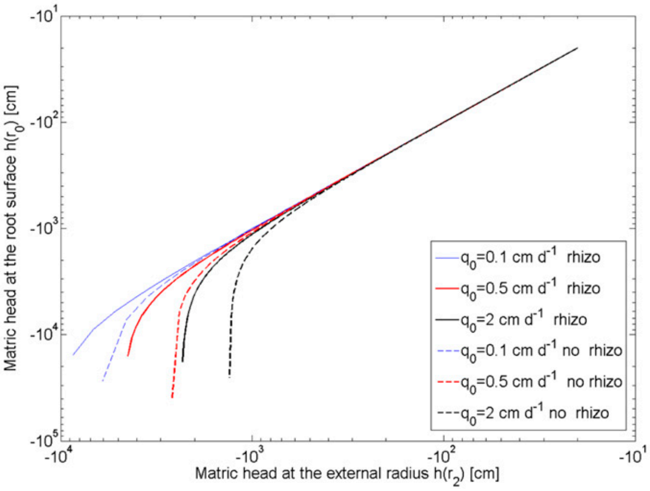

Fig. 7. Matric potential at the root surface $h(r_0,t)$ as a function of the matric potential at the outer radius of the domain $h(r_2,t)$, calculated for various uptake rates q_0, with a rhizosphere (solid lines) and with no rhizosphere (dashed lines).

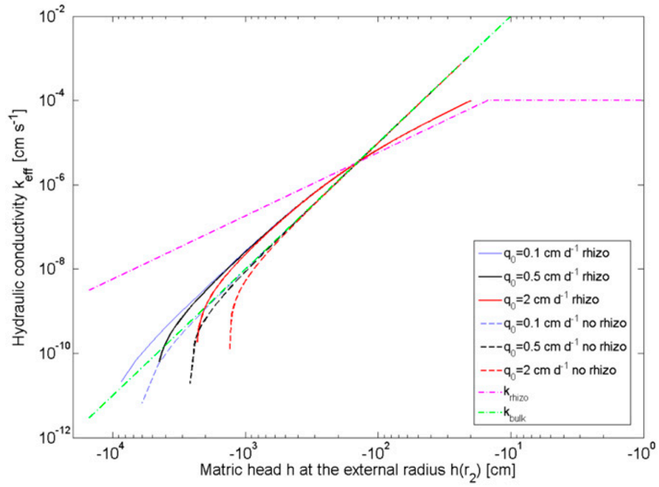

Fig. 8. Effective conductivity of the soil and rhizosphere, k_{eff}, calculated according to Eq. [9] and plotted for the cases with and without a rhizosphere for various uptake rates q_0. The results are compared with the actual conductivities of the rhizosphere and the bulk soil, $k_{\text{rhizo}}(h)$ and $k_{\text{bulk}}(h)$, respectively.

Appendix A. It is interesting to evaluate how $k_{\text{eff}}(h)$ changes as the soil becomes dry and compare it with the actual conductivities of the soil and rhizosphere, $k_{\text{bulk}}(h)$ and $k_{\text{rhizo}}(h)$, respectively. In the following, we consider $h(r_2,t)$ as the matric potential in the bulk soil. To be precise, the average matric potential in the bulk soil should be calculated at approximately $r = 0.6r_2$ (de Jong van Lier et al., 2006); however, our approximation does not change the general picture. In Fig. 8, we plotted k_{eff} as a function of $h(r_2)$ for the case with and without a rhizosphere for various q_0. Additionally, we compare the calculated $k_{\text{eff}}(h)$ with $k_{\text{rhizo}}(h)$ and $k_{\text{bulk}}(h)$.

Figure 8 shows that for the case with no rhizosphere, the calculated $k_{\text{eff}}(h)$ was equal to $k_{\text{bulk}}(h)$ for high matric potential. Precisely, for the low uptake rate, $k_{\text{eff}}(h) = k_{\text{bulk}}(h)$ for $h > -1000$ cm; for the high uptake rate, $k_{\text{eff}}(h) = k_{\text{bulk}}(h)$ for $h > -500$ cm. In this range of matric potentials, $k_{\text{eff}}(h)$ was not a function of the uptake rate. The water uptake can be considered as linear, in the sense that the flux was proportional to the matric potential gradient. For lower matric potentials, $k_{\text{eff}}(h)$ declined more rapidly than $k_{\text{bulk}}(h)$. The decrease in $k_{\text{eff}}(h)$ was more pronounced for higher uptake rates. The abrupt decrease in $k_{\text{eff}}(h)$ was caused by the drop in matric potential at the root surface. When the drop in matric potential occurred, the conductivity toward the roots decreased significantly and the process became strongly nonlinear.

In the case with a rhizosphere, for $h > -140$ cm, $k_{\text{eff}}(h)$ was slightly higher than $k_{\text{rhizo}}(h)$ and decreased as $k_{\text{rhizo}}(h)$ decreased. In this range of matric potentials, the rhizosphere was the limiting component and controlled $k_{\text{eff}}(h)$. For $h < -140$ cm, $k_{\text{eff}}(h)$ became smaller than $k_{\text{rhizo}}(h)$ and its slope was similar to that of $k_{\text{bulk}}(h)$. In fact, in this range of matric potentials, $k_{\text{rhizo}}(h) > k_{\text{bulk}}(h)$ and $k_{\text{eff}}(h)$ was controlled by the bulk soil. For lower matric potentials, $h < -1000$ cm, $k_{\text{eff}}(h)$ declined more rapidly than $k_{\text{bulk}}(h)$ because of the drop

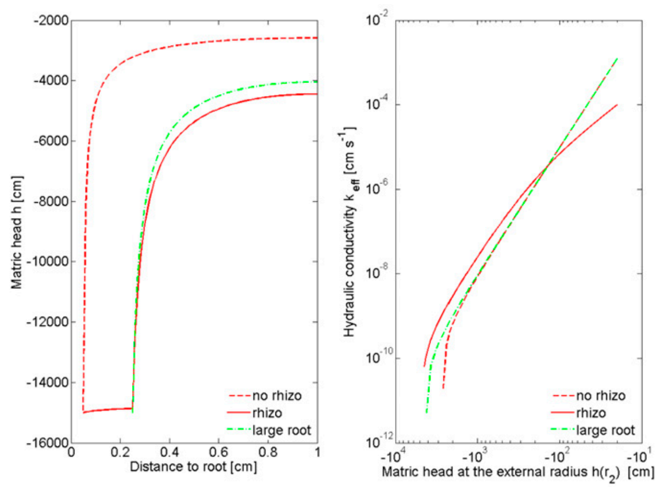

Fig. 9. The cases with and without a rhizosphere (solid and dotted red lines) compared with the ideal case of a root with radius equal to r_1 and with no rhizosphere, i.e., a large root (green line): matric potential profile at the limiting point (left) and effective conductivity of the flow to the root as calculated with Eq. [9] (right). The rhizosphere favors water availability to plants more than what a root with a larger radius would do.

in matric potential. Compared with the case with no rhizosphere, the effect of the rhizosphere was to lower k_{eff} for $h > -140$ cm and to increase k_{eff} for $h < -140$ cm. The relative increase in k_{eff} reached one to two orders of magnitude at low matric potentials.

Finally, we come to the question whether the effect of the rhizosphere can be simulated by simply extending the root radius. In Fig. 9, we compare the calculations with and without a rhizosphere and the calculations with a root having a radius of 0.25 cm and no rhizosphere (i.e., homogeneous bulk soil). The calculations were done with $r_2 = 1$ cm and $q_0 = 0.5$ cm d^{-1}. For the extended root radius, the flow rate at the root surface was 0.1 cm d^{-1}, so that the flow rate times the root radius was identical to the case with a rhizosphere. The calculations show that the extended root radius behaved like the rhizosphere in decreasing the bulk matric potential at the limiting point. In other words, extending the root radius increased the water availability, as reported already in de Jong van Lier et al. (2006); however, the effect of the rhizosphere was slightly larger than that of the extended root radius. The slight difference between the two cases is more evident in the effective hydraulic conductivity. Figure 9 (right) shows that $k_{\mathrm{eff}}(h)$ decreased less rapidly with the rhizosphere than with the extended root radius. The difference between the two cases was caused by the high soil water capacity of the rhizosphere at very low matric potentials. As the soil dried, the water uptake in the rhizosphere increased and compensated for the uptake in the bulk soil, therefore reducing the hydraulic gradients in the bulk soil. In conclusion, extending the effective root radius can be a "computational trick" for simulating a significant part of the behavior of the rhizosphere, but not all. Or the other way around, such a rhizosphere allows a root to take up the same or more water without having to grow in thickness.

Conclusions

The calculations showed that when the rhizosphere has a higher water holding capacity and unsaturated conductivity than the bulk soil, it favors root water uptake by attenuating the drop in water potential next to the roots. The drop in potential is alleviated due to the high unsaturated conductivity of the rhizosphere, as well as by the relatively reduced uptake in the bulk soil. Under these conditions, the rhizosphere works as an optimal hydraulic conductor as well as a reservoir of water that can be taken up when the water supply from the bulk soil becomes limiting.

The benefits for the plant are expected to be manifold. In terms of the volume of available water, the rhizosphere increases the soil water holding capacity under unsaturated conditions and decreases the average water content of the bulk soil at which the root–soil interface reaches the wilting point. The consequence is that a higher volume of water is available to plants before the wilting point at the root surface is reached. Similarly, the rhizosphere reduces the water potential gradients toward the roots when soils dry. Expressed in terms of conductivity, this means a relative increase in the effective conductivity of the soil–rhizosphere–plant system. This is likely to result in a relative increase in water potential inside the plant.

This study addressed one aspect of the complex and dynamic soil–plant interactions. Further studies are needed to elucidate the temporal dynamics of the rhizosphere properties during root growth, day–night cycles, and drying and irrigation events. The idea of this study, however, was to illustrate how the narrow region around the roots, with the supposed high capacity of retaining water during soil drainage, favors water flow to the roots. We expect that this mechanism has an impact at the plant scale and it should be included in macroscopic models of root water uptake. For instance, the rhizosphere can reduce the critical water content in the Feddes function (Feddes et al., 1976).

Appendix A

In the following, we provide the explicit solution of Eq. [7a–7f]. The solution is given in terms of the parameters a_i, b_i, and c_i, where the subscripts $i = 1,2$ refer to rhizosphere and bulk soil properties, respectively. To solve the system we make use of the relation

$$q=-\frac{\partial\phi}{\partial r}=-2ar-\frac{b}{r} \qquad \text{[A1]}$$

and we use Eq. [1], [2], and [4] to relate the state variables h and ϕ. We define

$$d_i\left(t\right)=-\frac{-h_{0,i}}{\lambda_i}\left[\frac{h\left(r_1,t\right)}{h_{0,i}}\right]^{\lambda_i+1},\quad i=1,2 \qquad \text{[A2]}$$

Then the parameters a_i, b_i, and c_i are

$$a_2\left(t\right)=\frac{0.5q_0r_0d_1\left(t\right)}{d_2\left(t\right)\left(r_1^2-r_0^2\right)-d_1\left(t\right)\left(r_1^2-r_2^2\right)} \qquad \text{[A3]}$$

$$a_1\left(t\right)=\frac{q_0r_0+2a_2\left(t\right)\left(r_1^2-r_2^2\right)}{2\left(r_1^2-r_0^2\right)} \qquad \text{[A4]}$$

$$b_1\left(t\right)=-q_0r_0-2a_1\left(t\right)r_0^2 \qquad \text{[A5]}$$

$$b_2\left(t\right)=-2a_2\left(t\right)r_2^2 \qquad \text{[A6]}$$

$$c_i\left(t\right)=-a_i\left(t\right)r_1^2-b_i\left(t\right)\ln r_1 + \frac{k_{0,i}\left(1-\tau_i\right)}{h_{0,i}^{-\tau_i}}h\left(r_1,t\right)^{1-\tau_i},\quad i=1,2 \qquad \text{[A7,8]}$$

where $h(r_1,t = 0) = h_{\mathrm{init}}$ and the increment of $h(r_1,t)$ is calculated as

$$\left.\frac{\partial h}{\partial t}\right|_{r_1}=\frac{4a_i\left(t\right)}{C_i\left[h\left(r_1,t\right)\right]} \qquad \text{[A9]}$$

We derived the effective conductivity k_{eff} (m s^{-1}) of the flow domain. We assumed that the soil conductivity k is constant across r. Because $q = -k(\partial h/\partial r)$, Eq. [A1] becomes

$$k\left(h\right)\frac{\partial h}{\partial r}=2ar+\frac{b}{r} \qquad \text{[A10]}$$

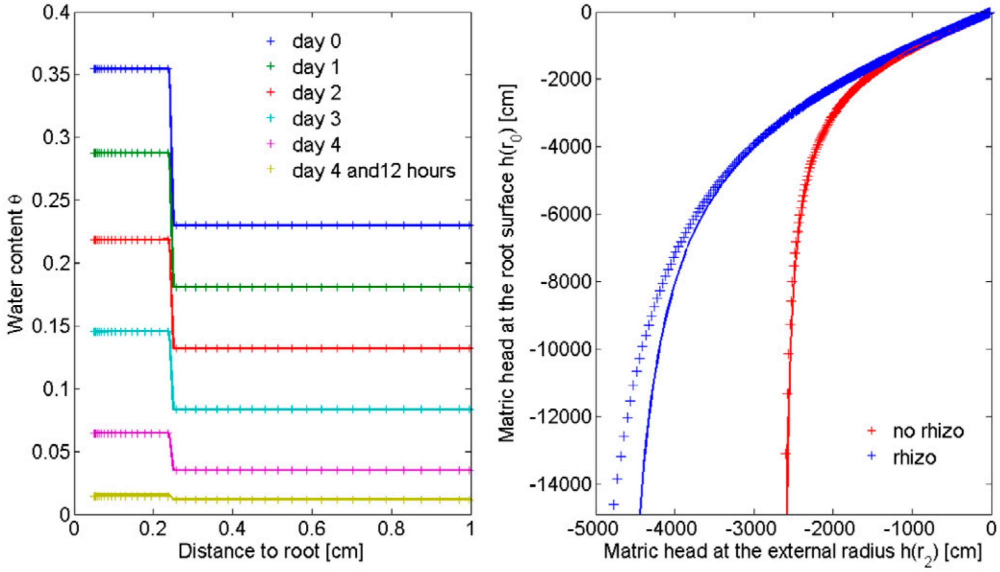

Fig. 10. Comparison between the numerical (crosses) and analytical solutions (solid lines) for water content profiles toward the root with time calculated for the case with a rhizosphere (left) and matric head at the root surface $h(r_0)$ vs. matric head the external radius $h(r_2)$ for the cases with and without a rhizosphere (right).

The differential Eq. [A9] can be solved by the separation of variables:

$$k(h)\,\mathrm{d}h = \left(2ar + \frac{b}{r}\right)\mathrm{d}r \qquad \text{[A11]}$$

Under the assumption that k is constant, integration over r_2 and r_0 gives

$$k\left[h(r_2) - h(r_0)\right] = a\left(r_2^2 - r_0^2\right) + b\ln\left(\frac{r_2}{r_0}\right) \qquad \text{[A12]}$$

Substitution of a and b computed for the case with no rhizosphere gives Eq. [9]. Note that Eq. [A12] is correct as long as $k[h(r_0,t)] \sim k[h(r_2,t)]$, i.e., for small gradients in matric potential toward the roots.

Appendix B

To validate the analytical solution, we wrote a numerical code according to the scheme presented by van Dam and Feddes (2000) and de Jong van Lier et al. (2006). The Richards equation in radial coordinates (Eq. [3]) was solved with the finite difference scheme of Eq. [11] and [12] in de Jong van Lier et al. (2006). The domain was discretized using a finer grid near the root and a larger grid at a greater distance from the root. The smallest grid size next to the root was set to 10^{-5} mm. We ran the numerical simulations for both cases, with a rhizosphere and with no rhizosphere, with identical boundary conditions, geometry, and hydraulic properties as in the analytical solutions.

The analytical and numerical solutions showed optimal agreement. The limiting point t_{lim} calculated with the analytical and numerical solutions differed by <0.1%. Interestingly, the error in t_{lim} was smaller in the case with a rhizosphere than in the case with no rhizosphere; this was probably due to the smaller gradients in water potentials occurring in the case with a rhizosphere. The water content profiles calculated for the case with a rhizosphere are plotted in Fig. 10 (left). The curves show an optimal match, although a small difference in water content in the rhizosphere developed approaching the limiting point. After 4 d and 12 h, the water content in the rhizosphere was 0.0157 according to the analytical solution and 0.0145 according to the numerical simulation. This slight difference between the two solutions is visible in Fig. 10 (right), where we plotted the matric potential at the root surface $h(r_0)$ vs. the matric potential at the outer bulk soil radius $h(r_2)$. Figure 10 (right) shows that for the case with no rhizosphere, the analytical and numerical solutions match perfectly. For the case with a rhizosphere, the numerical solution calculated a smaller drop in matric potential toward the root and a more linear behavior compared with the analytical solution; however, this difference does not change the general picture of this study.

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